BUI THANH LIEM

Full-Stack Developer

+084375255073 buithanhliem5073@gmail.com https://github.com/Bui-Thanh-Liem Ho Chi Minh City

PROFESSIONAL SUMMARY

A dedicated full-stack developer with over 2 years of practical experience building scalable and high-performance enterprise applications. Proficient in the **Node.js (NestJS, Express.js)** ecosystem for powerful back-end services and **React/Next.js** for highly interactive and SEO-friendly user interfaces. Passionate about delivering comprehensive products and constantly learning new technologies.

WORK EXPERIENCE (*~2 YEARS*)

SaaS Technology Company 11/2023 - 04/2026

Full-Stack Developer

- Over two years of hands-on experience in developing and maintaining modern web applications, evolving from a self-learning and working phase into a full-time Backend Developer position.

SKILLS

Frontend	ReactJS, NextJS, React-native (expo), Zustand, Redux, React Query,…
Backend	Node.js (NestJS, ExpressJS), TypeORM, JWT, OAuth2, RESTful API, BullMQ (Queue), Socket.io (Real-time), Base rabbitMQ (Message broker), …
Database & Caching	MySQL, MongoDB, Redis, ...
DevOps & Deployment	Docker, Docker Compose, Nginx, Ubuntu, VPS, base AWS, Mobile App Deployment (CHplay), CI/CD (github action)...
Security	Rate limiting, Brute-force protection, XSS/CSRF/SQL Injection mitigation,...
English	Good technical documentation reading skills.
Khác	Typescript, git, swagger, tool AI, on-premise server**,** scalable system design, performance optimization, high concurrency handling…

PROJECTS

Projects Centralized CRM & Sales Operations Ecosystem 07/2025 - 02/2026

Role: Full Stack Developer | Team size: 10

- **Database Design** & **Optimization:** Engineered complex database schemas for core entities including Customers, Note Sessions, and Invoice Management, ensuring data integrity and system scalability.
- **Access Control** & **Security:** Implemented a hybrid access control model combining dynamic RBAC with resource-based (ABAC-style) policies to secure sensitive customer and financial data across hierarchical departments.
- **Dashboard** & **Analytics Development:** Developed centralized dashboard features to monitor revenue, track employee performance, and manage lead sources in real-time.
- **Multi-channel Data Integration:** Built robust APIs and Webhooks to automate the collection and normalization of lead data from third-party platforms (Social Media, Web, and external providers) into the CRM.
- **Background Processing** & **Caching:** Integrated BullMQ and Redis to handle intensive background services—such as notifications and data synchronization—while implementing strategic caching to reduce database load and maintain API latency under 200ms.
- **Architectural Evolution:** Led the strategic migration from a Monolithic architecture to a Microservices ecosystem, decoupling the system into independent services: Gateway, Auth, HRM, CRM, and PM.

- **Event-Driven Communication:** Leveraged RabbitMQ as a message broker to implement an event-driven architecture, enhancing system throughput and ensuring eventual consistency across distributed services.
- **DevOps** & **Deployment:** Orchestrated production environments using **Docker and Nginx on Ubuntu VPS**, ensuring high availability and streamlined deployment workflows.
- **Tech Stack**:
  - **Backend:** NestJS, TypeScript, MySQL, RabbitMQ, Redis, BullMQ, ...
  - **Frontend:** Next.js, TailwindCSS, React-native, Zustand, ...
  - **Infrastructure** & **DevOps:** Docker, Nginx, Linux (Ubuntu), on-premise server....

Social Media Web Application 01/2026 - 03/2026

Personal Project

- **High-Concurrency Engineering:** Engineered a backend capable of handling **>10,000 req/s** for core interactions (likes) by implementing **write-behind caching** and **optimistic/pessimistic locking** to ensure data integrity under heavy load.
- **Real-time Communication:** Architected a low-latency real-time system for messaging and instant notifications using **Socket.io** and **Redis-backed message queuing**.
- **Cloud Infrastructure** & **DevOps:** Orchestrated a scalable **AWS ecosystem** (ECS, ECR, ALB, CloudFront) with automated **CI/CD pipelines** via GitHub Actions, optimizing global content delivery and deployment frequency.
- **Frontend Optimization:** Developed a seamless, responsive UI using **Radix UI** and **TailwindCSS**, leveraging **React Query** for efficient server-state management and reduced API overhead.
- **Tech Stack**:
  - **Backend:** Express.js, MongoDB, Redis, BullMQ, Socket.io, Gemini api, Telegram api, ...
  - **Frontend:** React.js, TypeScript, Radix UI, Zustand, React Query, ...
  - **Infrastructure** & **DevOps:** AWS (ECS, S3, CloudFront, Route 53), Docker, GitHub Actions, ...
- **Repo**: https://github.com/Bui-Thanh-Liem/social-be

Scalable & Reliable E-commerce Backend 01/2026 - present

Personal Project

- **Transactional Integrity** & **Concurrency:** Leveraged strong transaction isolation levels and implemented both optimistic and pessimistic locking mechanisms with TypeORM and PostgreSQL to ensure data consistency in critical operations (orders, inventory, and promotions) under high concurrency.
- **Caching for Performance:** Integrated Redis for write-through and read-through caching of frequently accessed data (products, inventories, sessions) to drastically reduce database load and latency during traffic spikes and flash sales.
- **Rich E-commerce Feature Set:** Delivered full suite of modules: user/auth management, product & inventory management, shopping cart, order lifecycle, campaign/voucher promotion, and multi-role access control.
- **Payment Integration:** Seamlessly incorporated multiple Vietnamese payment gateways (MoMo, VNPay, ZaloPay) with secure and extensible payment flows.
- **Scalable Cloud-native Infrastructure:** Adopted AWS S3 for media storage, with Docker-based deployment flows for portability and scalability.
- **API Design** & **Validation:** Built RESTful APIs with OpenAPI documentation, employing advanced input validation and error handling for robust integrations.
- **Tech Stack:**
  - **Backend**: NestJS, TypeScript, PostgreSQL, TypeORM, Redis, AWS S3, Passport (JWT, OAuth2), Multer, ...
  - **Infrastructure** & **DevOps**: Docker, GitHub Actions (CI/CD), AWS S3, ...
  - **Misc**: class-validator, winston (logging), RxJS. ...
- **Repo**: https://github.com/Bui-Thanh-Liem/ecommerce-be

CERTIFICATIONS

Information Technology Professional Diploma 04/2024

EDUCATION

Ho Chi Minh City Information Technology College (ITC) 2022 - 04/2024

Information Technology

Completed the academic program while gaining practical experience as a Backend Developer.

INTERESTS

Football and exploring new tech.